

Polynomials

$\mathbb{Z}[x]$ is the set of integer coefficient polynomials.

A polynomial $P \in \mathbb{Z}[x]$ is *irreducible* over $\mathbb{Z}$ if there do not exist non-constant polynomials $Q, R \in \mathbb{Z}[x]$ such that

$$P(x) = Q(x)R(x).$$

Problems

- **P1 (O1)** Show that $x^2 + 1$ is irreducible over $\mathbb{Z}$.
- **P2 (O1)** Find all primes p such that $x^p + 1$ is irreducible.
- **P3 (O1)** For which integers a is $x^2 + ax + 1$ irreducible over $\mathbb{Q}$?
- **P4 (O1)** Let $P \in \mathbb{Z}[x]$ be a polynomial that is not irreducible. Show that $P(n)$ is prime for only finitely many $n \in \mathbb{Z}$.
- **P5 (MAOL).** Does there exist a polynomial $P \in \mathbb{Z}[x]$ that does not have integer roots, such that for every prime p , there exists $n \in \mathbb{Z}$ with

$$p \mid P(n)?$$